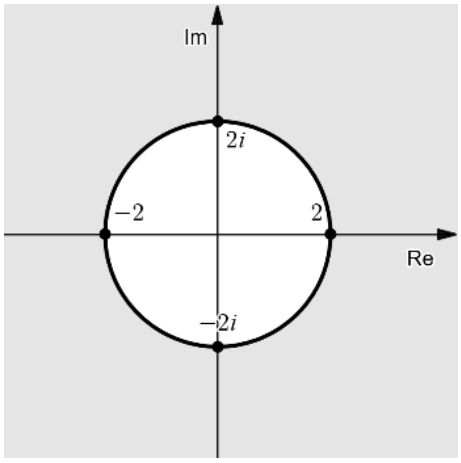
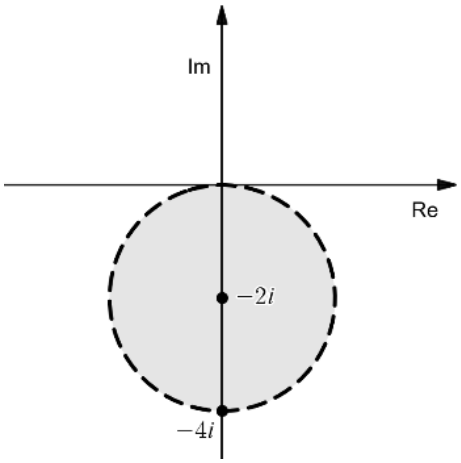
1) a)



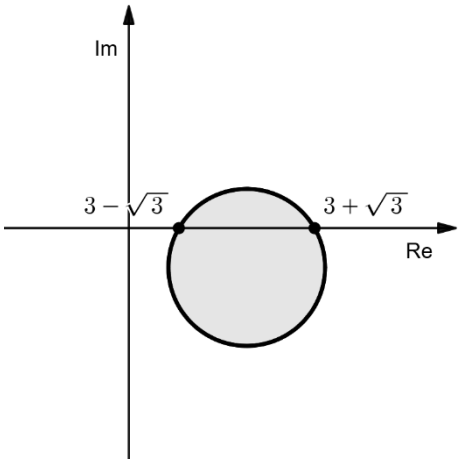
1) b)



1) c)

$(x-3)^2+(y+1)^2=4$

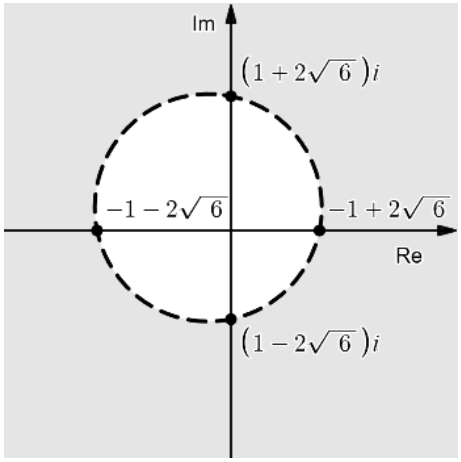
When $y=0, x=3\pm\sqrt{3}$



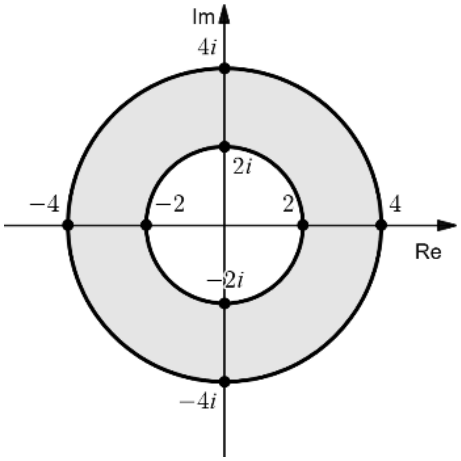
1) d)

$(x+1)^2+(y-1)^2=25$

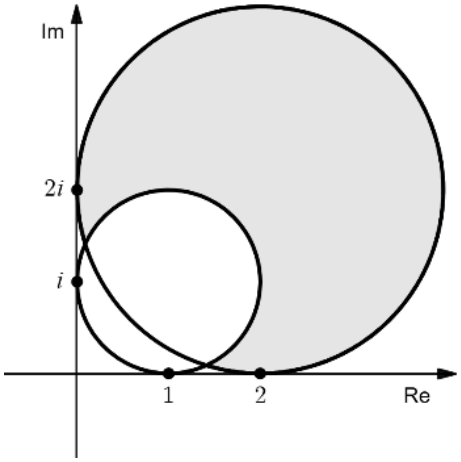
When $x=0$, $y=1\pm2\sqrt{6}$ and when $y=0$, $x=-1\pm2\sqrt{6}$



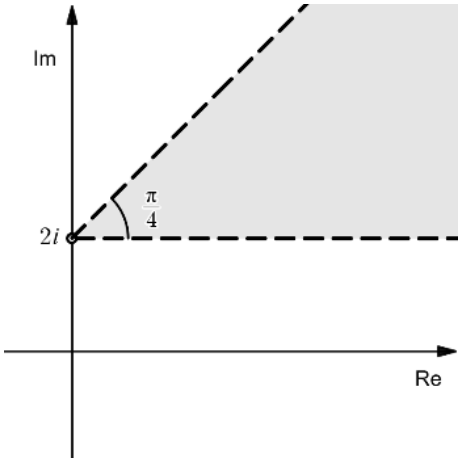
1) e)



1) f)



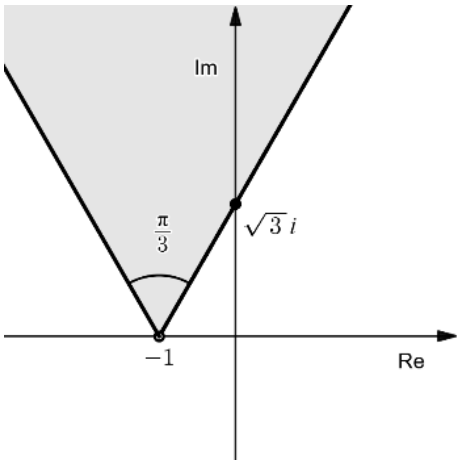
2) a)



2) b)

$y = \tan\frac{\pi}{3}(x+1) \Rightarrow y = \sqrt{3}x + \sqrt{3}$

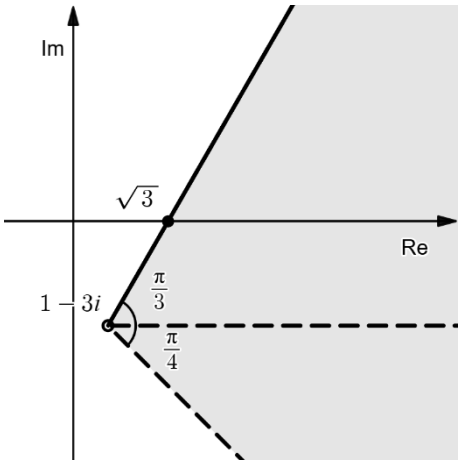
When $x = 0$, $y = \sqrt{3}$



2) c)

$y + 3 = \tan\frac{\pi}{3}(x - 1) \Rightarrow y = \sqrt{3}x - \sqrt{3} - 3$

When $y = 0$, $x = \frac{\sqrt{3}+3}{\sqrt{3}} = \sqrt{3}$



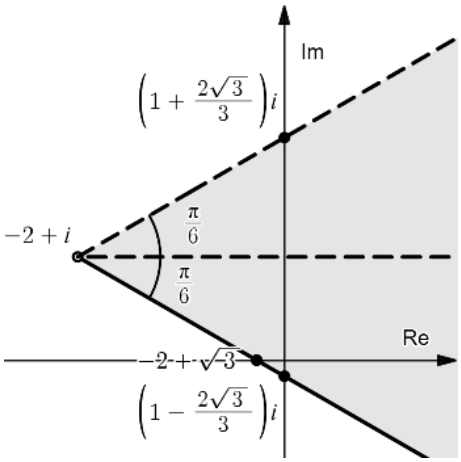
2) d)

$y - 1 = \tan\frac{\pi}{6}(x + 2) \Rightarrow y = \frac{\sqrt{3}}{3}x + \frac{2\sqrt{3}}{3} + 1$

When $x = 0$, $y = \frac{2\sqrt{3}}{3} + 1$

$y - 1 = \tan\left(-\frac{\pi}{6}\right)(x + 2) \Rightarrow y = -\frac{\sqrt{3}}{3}x - \frac{2\sqrt{3}}{3} + 1$

When $x = 0$, $y = -\frac{2\sqrt{3}}{3} + 1$ and when $y = 0$, $x = \frac{-\frac{2\sqrt{3}}{3}+1}{\frac{\sqrt{3}}{3}} = 2 - \sqrt{3}$



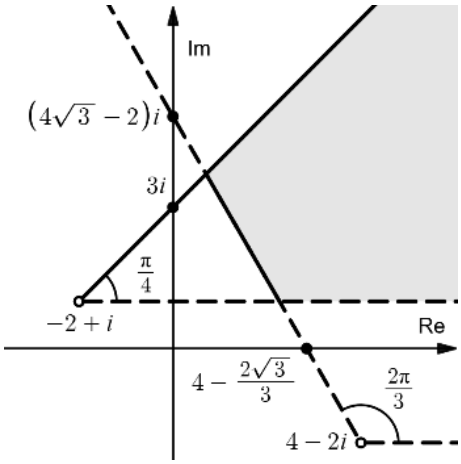
2) e)

$y - 1 = \tan\frac{\pi}{4}(x + 2) \Rightarrow y = x + 3$

When $x = 0$, $y = 3$

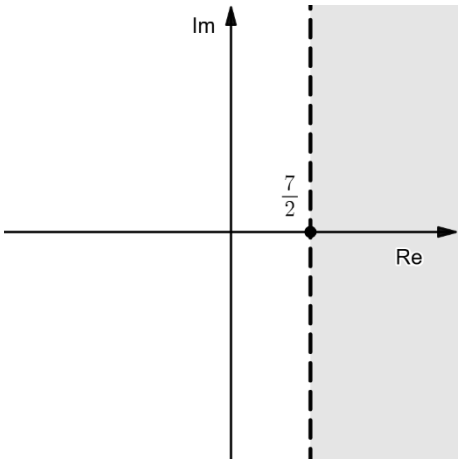
$y + 2 = \tan\left(\frac{2\pi}{3}\right)(x - 4) \Rightarrow y = -\sqrt{3}x + 4\sqrt{3} - 2$

When $x = 0$, $y = -2$ and when $y = 0$, $x = \frac{-4\sqrt{3}+2}{-\sqrt{3}} = 4 - \frac{\sqrt{3}}{2}$



3) a)

Midpoint: $(7, 2)$

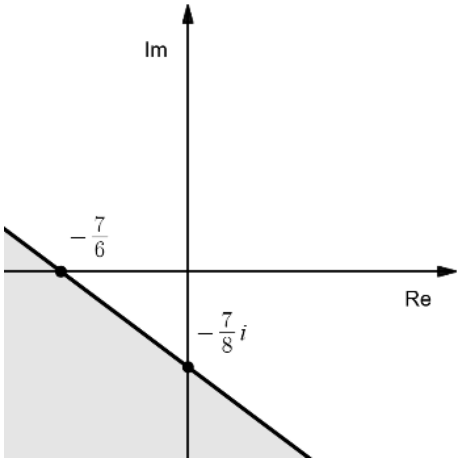


3) b)

Midpoint: $\left(\frac{3}{2}, -2\right)$ and Gradient: $\frac{4}{3}$

Line: $y + 2 = -\frac{3}{4}\left(x - \frac{3}{2}\right) \Rightarrow y = -\frac{3}{4}x - \frac{7}{8}$

When $x = 0$, $y = -\frac{7}{8}$ and when $y = 0$, $x = -\frac{7}{6}$

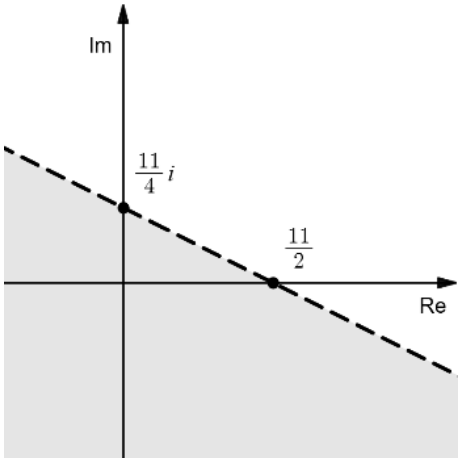


3) c)

Midpoint: $\left(-\frac{1}{2}, 3\right)$ and Gradient: 2

Line: $y - 3 = -\frac{1}{2}\left(x + \frac{1}{2}\right) \Rightarrow y = -\frac{1}{2}x + \frac{11}{4}$

When $x = 0$, $y = \frac{11}{4}$ and when $y = 0$, $x = \frac{11}{2}$

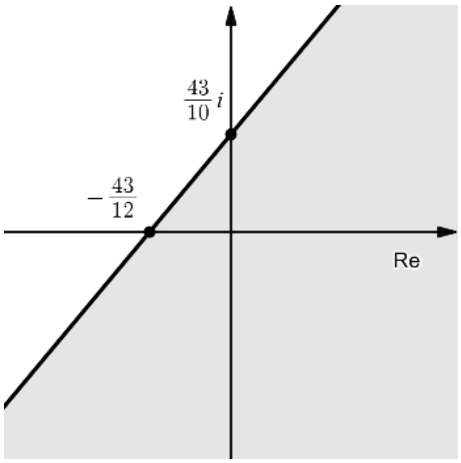


3) d)

Midpoint: $\left(1,\frac{11}{2}\right)$ and Gradient: $-\frac{5}{6}$

Line: $y-\frac{11}{2}=\frac{6}{5}(x-1) \Rightarrow y=\frac{6}{5}x+\frac{43}{10}$

When $x=0$, $y=\frac{43}{10}$ and when $y=0$, $x=-\frac{43}{12}$



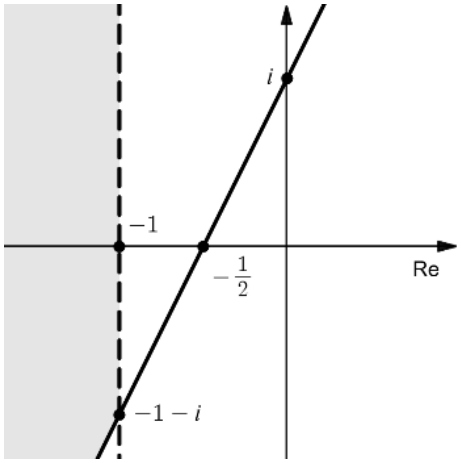
3) e)

Midpoint: $(-1,0)$ gives $x=-1$

Midpoint: $(1,3)$ and Gradient: $-\frac{1}{2}$

Line: $y-3=2(x-1) \Rightarrow y=2x+1$

When $x=0$, $y=1$ and when $y=0$, $x=-\frac{1}{2}$



4) a)

$|z-2-2\mathrm{i}|=2$

$\tan\left(\frac{1}{1}\right)=\frac{\pi}{4}$ which gives $\arg(z-2-2\mathrm{i})=\frac{\pi}{4}$

$\arg(z-2-2\mathrm{i})\leq\frac{\pi}{4}$ and $|z-2-2\mathrm{i}|\leq 2$

$\left\{z\in\mathbb{C}:\arg(z-2-2\mathrm{i})\leq\frac{\pi}{4}\right\}\cap\{z\in\mathbb{C}:|z-2-2\mathrm{i}|\leq 2\}$

4) b)

$|z-1|=|z-3|$

$|z-2|=1$

$|z-2|<1$ and $|z-1|\leq|z-3|$

$\{z\in\mathbb{C}:|z-2|<1\}\cap\{z\in\mathbb{C}:|z-1|\leq|z-3|\}$

4) c)

$|z-2\mathrm{i}|=|z-2|$

$\tan\left(\frac{1}{1}\right)=\frac{\pi}{4}$ which gives $\arg(z-2)=\frac{\pi}{4}$ and $\arg(z-2)=\frac{3\pi}{4}$

$|z-2\mathrm{i}|>|z-2|$ and $\frac{\pi}{4}\leq\arg(z-2)<\frac{3\pi}{4}$

$\{z\in\mathbb{C}:|z-2\mathrm{i}|>|z-2|\}\cap\left\{z\in\mathbb{C}:\frac{\pi}{4}\leq\arg(z-2)<\frac{3\pi}{4}\right\}$

4) d)

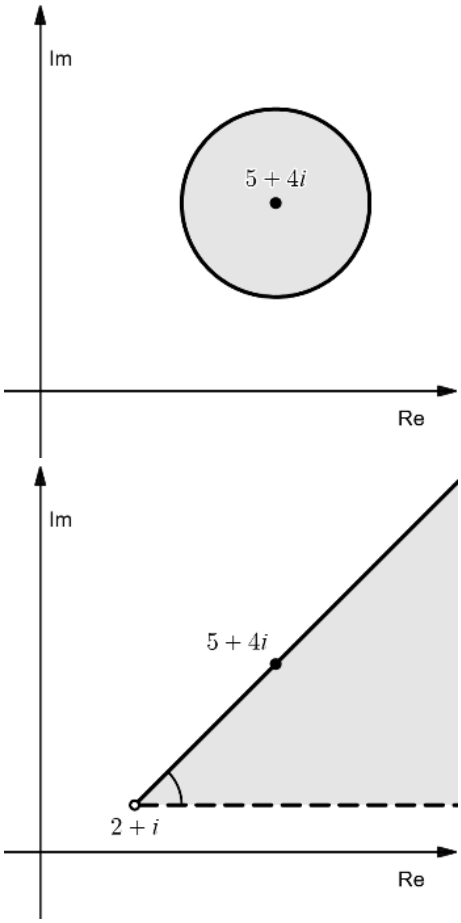
$|z-1-\mathrm{i}|=2$

$|z-1-\mathrm{i}|=1$

$|z-1-\mathrm{i}|\leq 2$ and $|z-1-\mathrm{i}|>1$

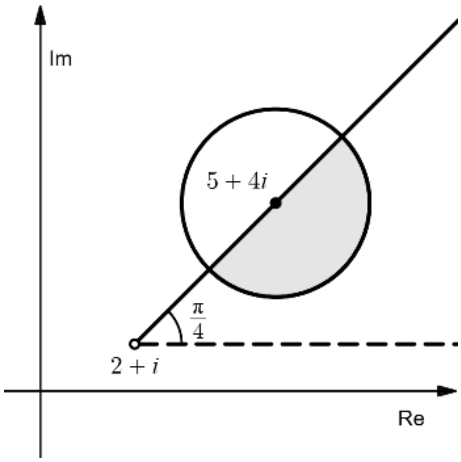
$\{z\in\mathbb{C}:1<|z-1-\mathrm{i}|\leq 2\}$

5) a)

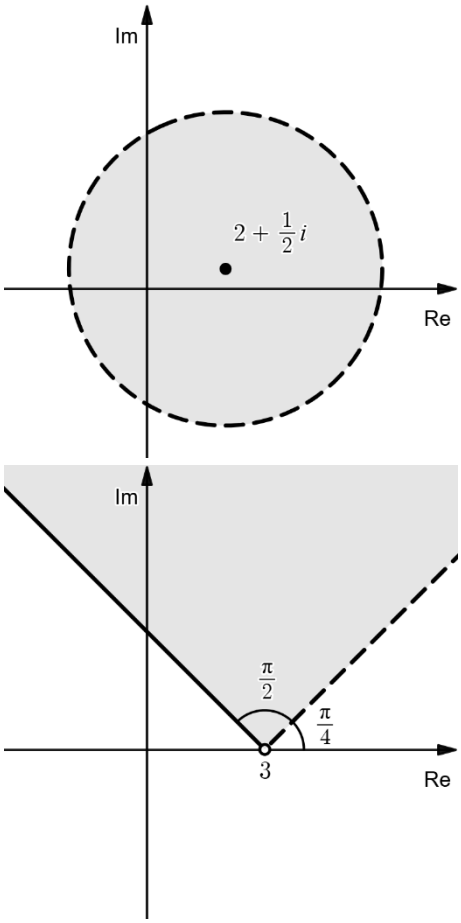


5) b)

$y - 1 = \tan\frac{\pi}{4}(x - 2) \Rightarrow y = x - 1$ so the line passes through $(5, 4)$



6) a)



6) b)

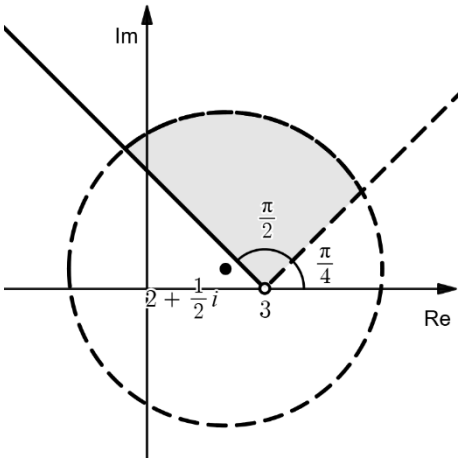
Consider the $\arg(z-3)=\frac{3\pi}{4}$ in cartesian form:

$y=\tan\left(\frac{3\pi}{4}\right)(x-3) \Rightarrow y=-(x-3)=-x+3$

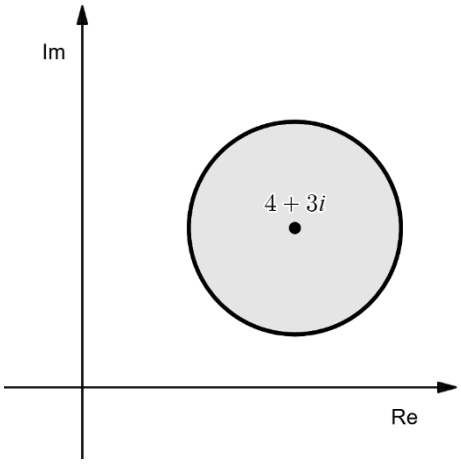
When $x=2$, $y=-2+3=1$ so the line lies above the point $2+\frac{1}{2}\mathrm{i}$

So $2+\frac{1}{2}\mathrm{i}$ is not in the region.

6) c)



7) a)



7) b)

Find where $y = mx$ meets the equation of the circle $(x+4)^2 + (y+4)^2 = 4$

$(x-4)^2 + (mx-3)^2 = 4$

$x^2 - 8x + 16 + m^2x^2 - 6mx + 9 - 4 = 0$

$(1+m^2)x^2 + (-8-6m)x + 21 = 0$

Consider $b^2 - 4ac = 0$:

$(-8-6m)^2 - 4 \times 21(1+m^2) = 0$

$36m^2 + 96m + 64 - 84 - 84m^2 = 0$

$-48m^2 + 96m - 20 = 0$

$12m^2 - 24m + 5 = 0$

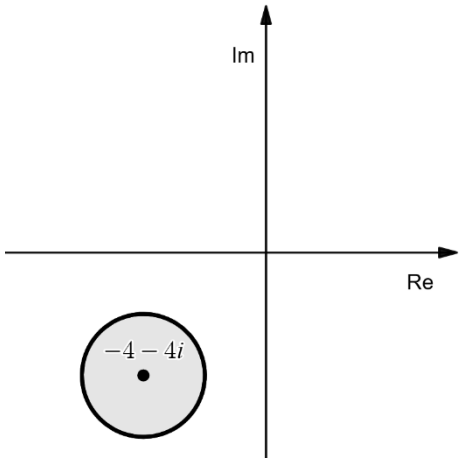
$m = \frac{24 \pm \sqrt{576-240}}{24} = 1 \pm \frac{\sqrt{21}}{6}$

$m = \tan\theta \therefore \arctan\left(1+\frac{\sqrt{21}}{6}\right) = 1.055017\ldots$ and $\arctan\left(1-\frac{\sqrt{21}}{6}\right) = 0.231984\ldots$

Therefore the least and greatest value of $\arg z$ are:

0.232 (3 SF) and 1.06 (3 SF)

8) a)



8) b)

Find where $y = mx$ meets the equation of the circle $(x+4)^2 + (y+4)^2 = 4$

$(x+4)^2 + (mx+4)^2 = 4$

$x^2 + 8x + 16 + m^2x^2 + 8mx + 16 - 4 = 0$

$(1+m^2)x^2 + (8+8m)x + 28 = 0$

Consider $b^2 - 4ac = 0$:

$(8+8m)^2 - 4 \times 28(1+m^2) = 0$

$64m^2 + 128m + 64 - 112 - 112m^2 = 0$

$3m^2 - 8m + 3 = 0$

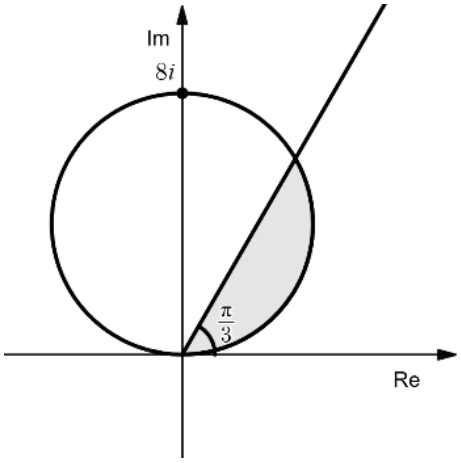
$m = \frac{8 \pm \sqrt{64-36}}{6} = \frac{4}{3} \pm \frac{\sqrt{7}}{3}$

$m = \tan\theta \therefore \arctan\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right) = 1.146765\ldots$ and $\arctan\left(\frac{4}{3} - \frac{\sqrt{7}}{3}\right) = 0.424031\ldots$

Therefore the least and greatest value of $\arg z$ are:

$1.146765\ldots - \pi = -1.99$ (3 SF) and $0.424031\ldots - \pi = -2.72$ (3 SF)

9) a)



Maximum value of $|z|$ is the point where the circle and the line intersect:

$x^2 + (y-4)^2 = 16$ and $y = \tan\left(\frac{\pi}{3}\right)x = \sqrt{3}x$

$x^2 + \left(\sqrt{3}x - 4\right)^2 = 16$

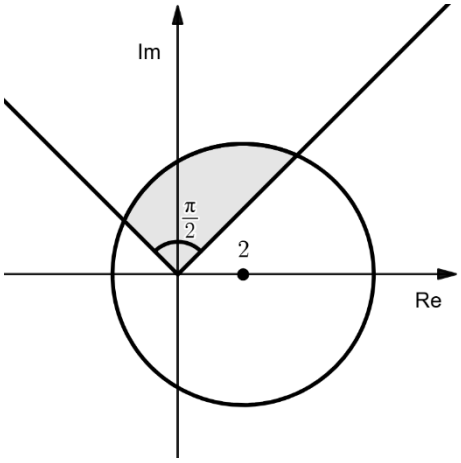
$x^2 + 3x^2 - 8\sqrt{3}x + 16 = 16$

$4x^2 - 8\sqrt{3}x = 0$

$4x\left(x - 2\sqrt{3}\right) = 0 \therefore x = 2\sqrt{3}, y = 6$

Maximum value of $|z| = \sqrt{\left(2\sqrt{3}\right)^2 + 6^2} = 4\sqrt{3}$

9) b)



Maximum value of $|z|$ is the point where the circle and the line created by $\arg z = \frac{\pi}{4}$ intersect:

$(x-2)^2 + y^2 = 16$ and $y = \tan\left(\frac{\pi}{2}\right)x = x$

$(x-2)^2 + x^2 = 16$

$x^2 - 4x + 4 + x^2 = 16$

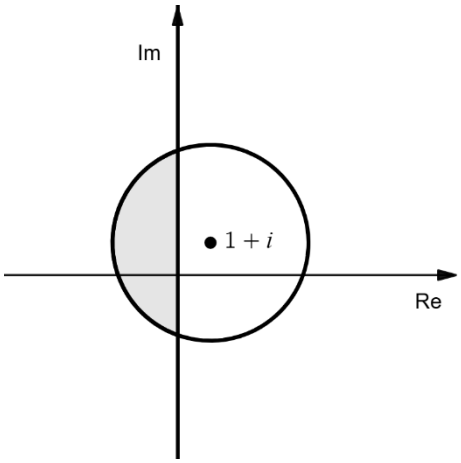
$2x^2 - 4x - 12 = 0 \ \ \therefore x = 1 \pm \sqrt{7}, y = 1 \pm \sqrt{7}$

Maximum value of $|z| = \sqrt{\left(1+\sqrt{7}\right)^2 + \left(1+\sqrt{7}\right)^2}$

$= \sqrt{16 + 4\sqrt{7}} = \sqrt{4\left(4+\sqrt{7}\right)} = 2\sqrt{4+\sqrt{7}}$

9) c)

Midpoint $(0,1)$ so line is $x = 0$



Maximum value of $|z|$ is the point where the circle crosses the imaginary axis:

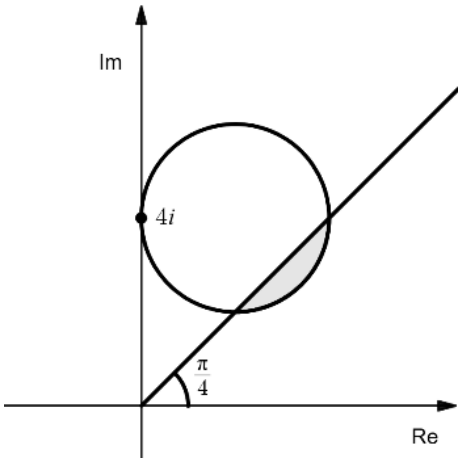
$(x-1)^2 + (y-1)^2 = 9$ and $x = 0$

$(y-1)^2 = 8$

$\therefore y = 1 \pm 2\sqrt{2}, x = 0$

Maximum value of $|z| = \sqrt{0^2 + \left(1 + 2\sqrt{2}\right)^2} = 1 + 2\sqrt{2}$

10) a)



Minimum value of $|z|$ is the first point where the circle and the line intersect:

$(x-2)^2 + (y-4)^2 = 4$ and $y = \tan\left(\frac{\pi}{4}\right)x = x$

$(x-2)^2 + (x-4)^2 = 4$

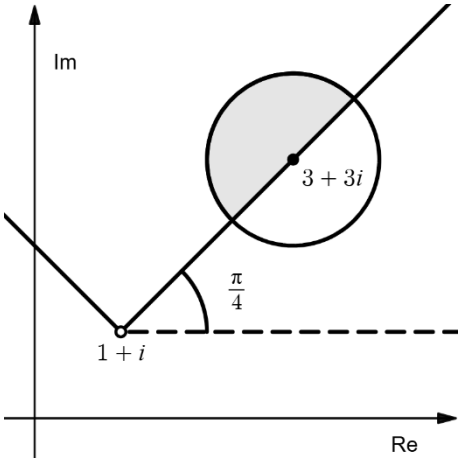
$x^2 - 4x + 4 + x^2 - 8x + 16 = 4$

$2x^2 - 12x + 16 = 0$

$\therefore x = 2$ or $x = 4$, so $(2,2)$ and $(4,4)$

Minimum value of $|z| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$

10) b)



Minimum value of $|z|$ is the first point where the $\arg z = \frac{\pi}{4}$ meets the circle:

$(x-3)^2 + (y-3)^2 = 2$ and $y = \tan\left(\frac{\pi}{4}\right)x = x$

$(x-3)^2 + (x-3)^2 = 2 \Rightarrow\ 2(x-3)^2 = 2 \Rightarrow\ (x-3)^2 = 1$

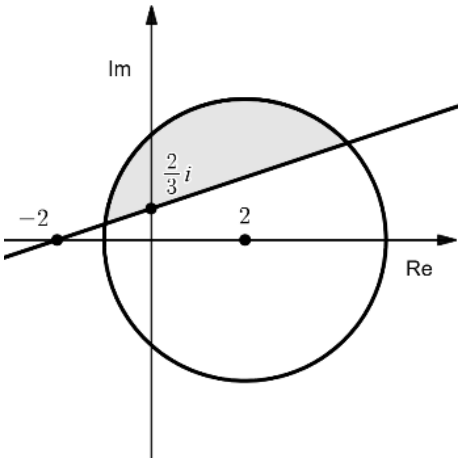
$\therefore x = 2$ or $x = 4$, so $(2,2)$ and $(4,4)$

Minimum value of $|z| = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2}$

10) c)

Midpoint: $(-5,-1)$ Gradient: -3

$y + 1 = \frac{1}{3}(x+5) \Rightarrow y = \frac{1}{3}x + \frac{2}{3}$



Minimum value of $|z|$ is the shortest distance to the line:

Find the perpendicular line passing through $(0,0)$: $y = -3x$

Find the point where these two lines intersect: $y = \frac{1}{3}x + \frac{2}{3}$ and $y = -3x$

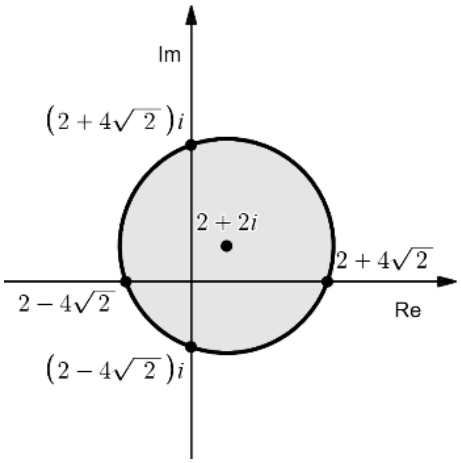
$\frac{1}{3}x + \frac{2}{3} = -3x \therefore x = -\frac{1}{5}$, $y = \frac{3}{5}$ so minimum value of $|z| = \sqrt{\left(-\frac{1}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \frac{\sqrt{10}}{5}$

11) a)

$(x-2)^2+(y-2)^2=36$

When $x=0$, $(y-1)^2=32 \Rightarrow y=2\pm 4\sqrt{2}$

When $y=0$, $(x-2)^2=35 \Rightarrow x=2\pm 4\sqrt{2}$



11) b)

$\left|9\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)\right|=9$

Distance to the maximum $|z|$ with angle $\frac{\pi}{4}$ is $\sqrt{2^2+2^2}+6=6+2\sqrt{2}$

Since $6+2\sqrt{2}<9$ then $9\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$ is outside the region.

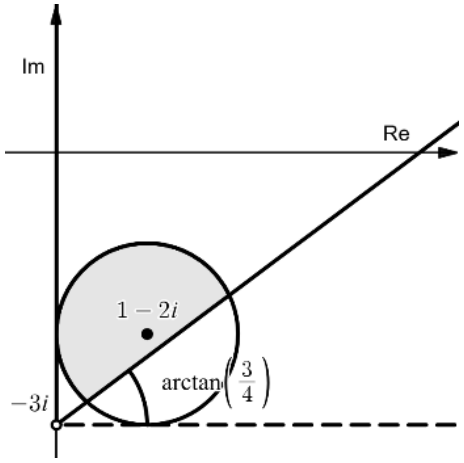
11) c)

$\left|4\left(\cos\frac{3\pi}{4}+\mathrm{i}\sin\frac{3\pi}{4}\right)\right|=4$

Distance away from when $\mathrm{Re}(z)=0$ is $\left|4\cos\frac{3\pi}{4}\right|=2\sqrt{2}$

Since $\left|2\sqrt{2}\right|<\left|2-4\sqrt{2}\right|$ then $4\left(\cos\frac{3\pi}{4}+\mathrm{i}\sin\frac{3\pi}{4}\right)$ lies inside the region.

12) a)



12) b)

Minimum value of $|z|$ is the shortest distance to the circle along the line which passes through the origin and $1 - 2i$:

$(x-1)^2 + (y+2)^2 = 1$ and $y = -2x$

$(x-1)^2 + (-2x+2)^2 = 1$

$x^2 - 2x + 1 + 4x^2 - 8x + 4 = 1$

$5x^2 - 10x + 4 = 0$

$\therefore x = 1 \pm \frac{\sqrt{5}}{5}, y = -2 \mp \frac{2\sqrt{5}}{5}$

so minimum value of $|z| = \sqrt{\left(1 - \frac{\sqrt{5}}{5}\right)^2 + \left(-2 + \frac{2\sqrt{5}}{5}\right)^2} = \sqrt{6 - 2\sqrt{5}} = \sqrt{5} - 1$

≈ 1.23 (3 SF)

13) a)

The circle $x^2 + y^2 = 25$ and the line $y = \tan\left(\frac{\pi}{3}\right)x = \sqrt{3}x$ when:

$x^2 + \left(\sqrt{3}x\right)^2 = 25$

$x^2 + 3x^2 = 25$

$x^2 = \frac{25}{4}$

$x = \pm\frac{5}{2}$ so $x = \frac{5}{2}$ and $y = \frac{5\sqrt{3}}{2}$

The circle $x^2 + y^2 = 25$ and the line $y = \tan\left(\frac{3\pi}{4}\right)x = -x$ when:

$x^2 + (-x)^2 = 25$

$2x^2 = 25$

$x^2 = \frac{25}{2}$

$x = \pm\sqrt{\frac{25}{2}}$ so $x = -\frac{5\sqrt{2}}{2}$ and $y = \frac{5\sqrt{2}}{2}$

So A is $-\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}\mathrm{i}$ and B is $\frac{5}{2} + \frac{5\sqrt{3}}{2}\mathrm{i}$

13) b)

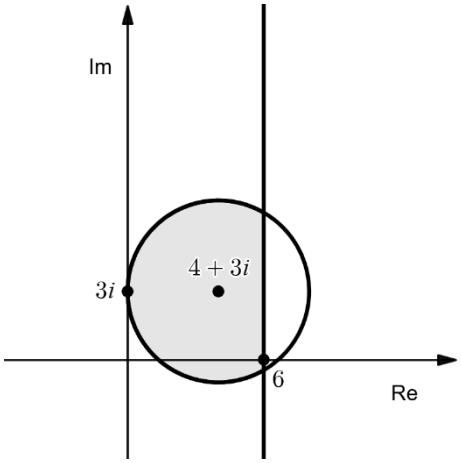
Area of sector $= \frac{1}{2} \times 5^2 \times \left(\frac{3\pi}{4} - \frac{\pi}{3}\right) = \frac{125\pi}{24}$ units^2

Area of circle $= 5^2\pi = 25\pi$ units^2

Shaded area $= 25\pi - \frac{125\pi}{24} = \frac{475\pi}{24}$ units^2

14) a)

Midpoint: $(6, 0)$

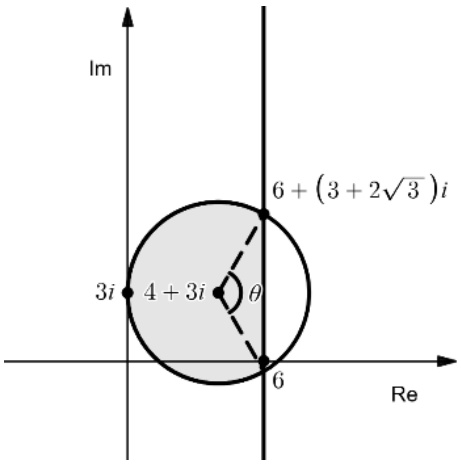


14) b)

Radius of circle: 4

Cartesian equation of circle: $(x-4)^2+(y-3)^2=16$

So when $x=6$, $y=3\pm 2\sqrt{3}$



Let $\phi=\frac{1}{2}\theta$

$\tan\phi=\frac{3+2\sqrt{3}-3}{6-4}=\sqrt{3}$

Angle: $\theta=2\times\arctan\sqrt{3}=2\arctan(\sqrt{3})=\frac{2\pi}{3}$

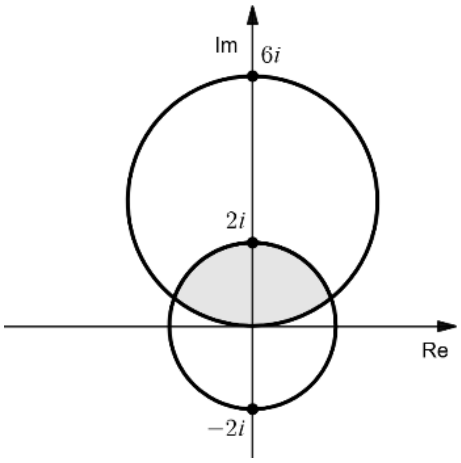
Area is Circle – Segment

Area of segment: $\frac{1}{2}r^2(\theta-\sin\theta)=\frac{1}{2}\times 4^2\left(\frac{2\pi}{3}-\sin\frac{2\pi}{3}\right)=\frac{16\pi}{3}-4\sqrt{3}$

Area of circle: $\pi r^2=16\pi$

Area of shaded region: $16\pi-\frac{16\pi}{3}+4\sqrt{3}=\frac{32\pi}{3}+4\sqrt{3}$ as req.

15)

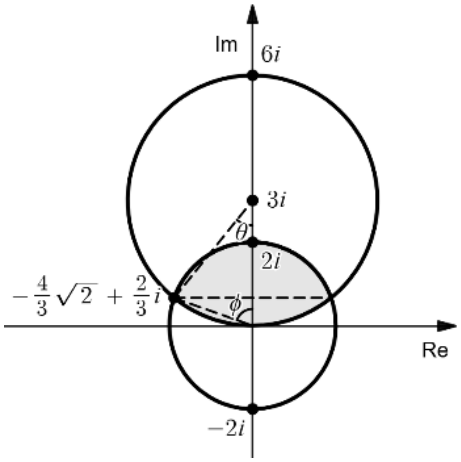


Points of intersection are where $x^2 + (y-3)^2 = 9$ and $x^2 + y^2 = 4$ intersect:

$4 - y^2 + (y-3)^2 = 9$

$4 - y^2 + y^2 - 6y + 9 = 9$

$\therefore y = \frac{2}{3}$ so $\left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}\right)$ and $\left(\frac{4\sqrt{2}}{3}, \frac{2}{3}\right)$



$\tan\theta = \frac{\frac{4\sqrt{2}}{3}}{3-\frac{2}{3}} = \frac{4\sqrt{2}}{7}$ $\therefore$ the required angle is $2\arctan\left(\frac{4\sqrt{2}}{7}\right)$

$\therefore$ segment of larger circle is

$\frac{1}{2} \times 3^2 \times 2\arctan\left(\frac{4\sqrt{2}}{7}\right) - \frac{1}{2} \times 3^2 \sin\left(2\arctan\left(\frac{4\sqrt{2}}{7}\right)\right) = 1.7172888\ldots$

$\tan\phi = \frac{\frac{4\sqrt{2}}{3}}{\frac{2}{3}} = 2\sqrt{2}$ $\therefore$ the required angle is $2\arctan\left(2\sqrt{2}\right)$

$\therefore$ segment of smaller circle is

$\frac{1}{2} \times 2^2 \times 2\arctan\left(2\sqrt{2}\right) - \frac{1}{2} \times 2^2 \sin\left(2\arctan\left(2\sqrt{2}\right)\right) = 0.5763007515\ldots$

Total is area is $\approx$ 2.29 units^2 (3 SF)